

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Speed: HDDs read and write around 80-160 MB/s due to spinning mechanical platters. SATA SSD read/write around 550 MB/s. NVMe SSD use PCIe lanes directly can reach 3000-7000 MB/s

Technology: HDDs store data magnetically on spinning disks with a moving read/write head, which can cause delay and with physical wear. SSDs (both SATA and NVMe) use flash memory chips with no moving parts, so they're faster and more durable.

Cost and capacity: HDDs offer the lowest cost per gigabyte, making them better for bulk storage. NVMe has the most cost per gigabyte but delivers the best performance, with SATA SSDs in between on both price and speed.

Usage:

HDD: HDD can be used for archiving large media libraries or backups — at places where capacity matters more than speed, due to HDDs' low cost per gigabyte.

SATA SSD: General OS boot drive for everyday computing (web browsing, office apps) — noticeably faster than HDD for boot times and app loads.

NVMe SSD: Video editing software like Adobe Premiere Pro, where editing 4K/8K footage and where high read/write speeds is required and only NVMe can deliver.